

abstract-audit: figure sources

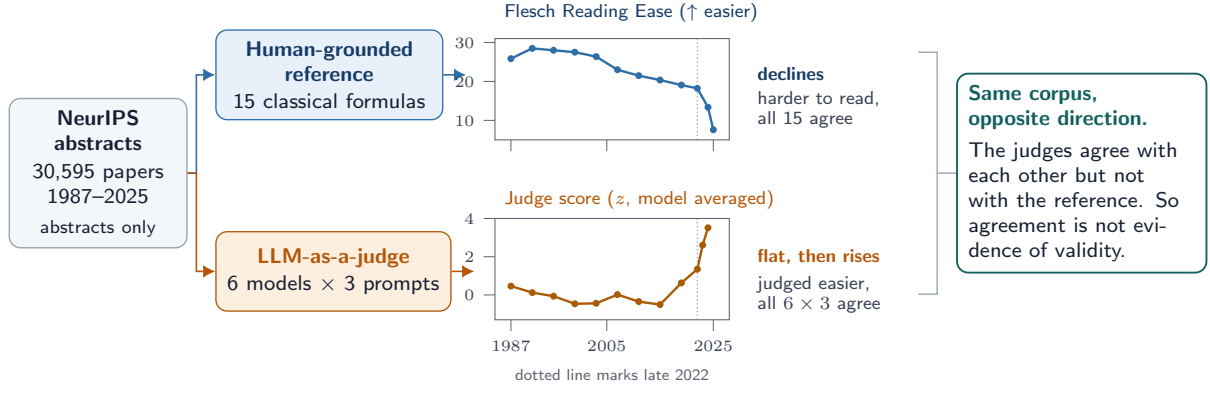


Figure 1: **Overview of the setup.** One corpus of 30,595 NeurIPS abstracts (1987–2025) is scored along two independent paths, 15 classical readability formulas calibrated on human reading performance, and six open-weight instruction-tuned models under three prompt templates, standardized per model. The reference falls (reading gets harder) while the judges are flat and then rise after 2022 (text is judged easier).

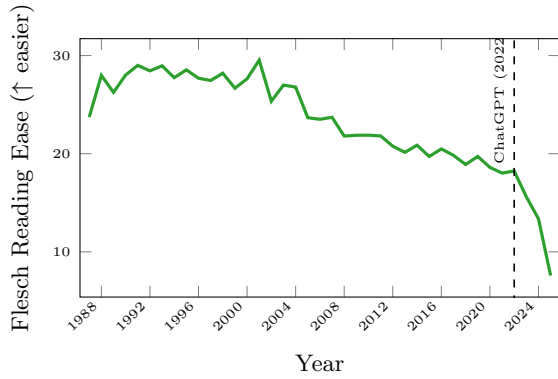


Figure 2: **Flesch Reading Ease on NeurIPS abstracts, 1987–2025.** Higher values indicate easier reading ($\uparrow$). NeurIPS falls from approximately 24 in 1987 to approximately 8 in 2025, a sustained decline. All 15 classical metrics move in the same direction on NeurIPS (Appendix ??). Vertical dashed line marks the late-2022 availability of widely-used instruction-tuned writing assistants.

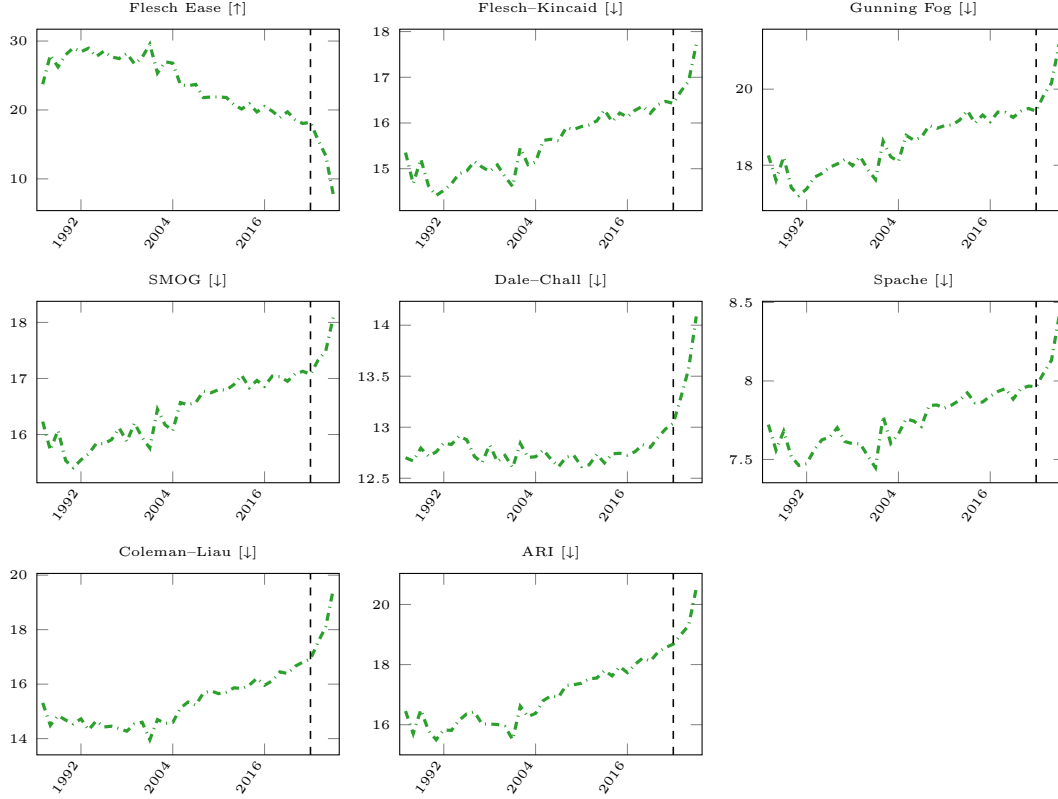


Figure 3: **Classical readability metrics on NeurIPS, part A: Flesch family and grade-level formulas, 1987–2025.** Arrows indicate the direction of easier reading ($\uparrow$ = higher is better; $\downarrow$ = lower is better). Every grade-level metric rises on NeurIPS after 2015 and accelerates after 2020, and Flesch Reading Ease drops correspondingly. Vertical dashed lines mark the late-2022 availability of instruction-tuned writing assistants. Part B with the remaining seven metrics appears in Figure 4.

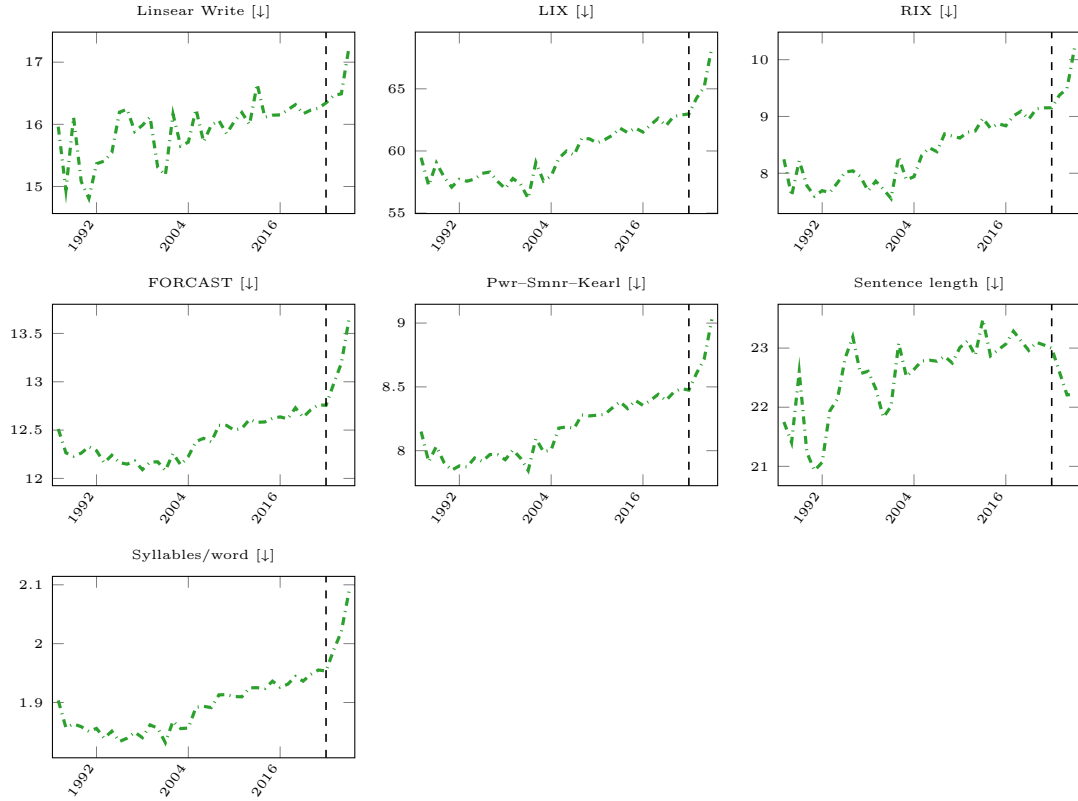


Figure 4: **Classical readability metrics on NeurIPS, part B: Linsear Write, LIX, RIX, FORCAST, Powers–Sumner–Kearl, sentence length, and syllables per word, 1987–2025.** Arrows indicate the direction of easier reading ($\downarrow$ = lower is better). Vertical dashed lines mark the late-2022 availability of instruction-tuned writing assistants. Part A with the Flesch family and the first eight grade-level formulas appears in Figure 3.

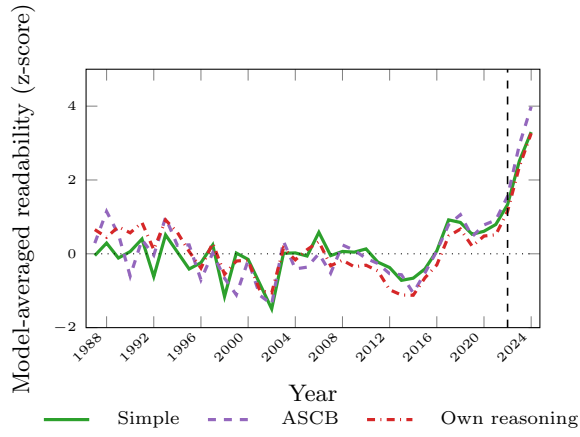


Figure 5: **LLM as judge readability on NeurIPS abstracts, 1987–2024.** Each curve is the mean z-score across all six open-weight models for one prompt variant, standardized to each model’s own 1987–2022 baseline (dotted line at zero). Under all three prompts the judged readability is approximately flat before 2022 and rises after the availability of instruction-tuned writing assistants (vertical dashed line). The direction is opposite to the classical readability decline in Figure 2.

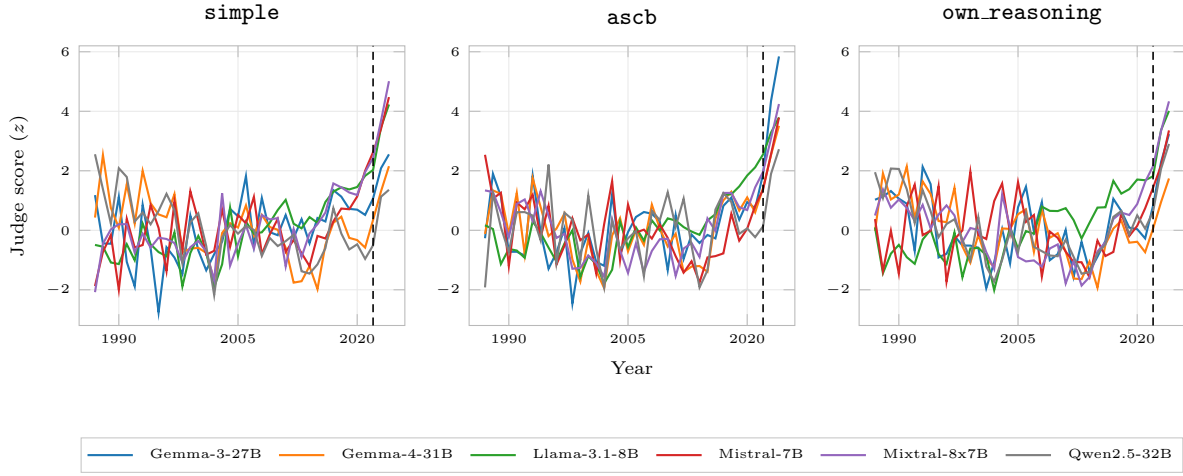


Figure 6: **Per-model judge scores by prompt template, standardized separately.** Each panel is one prompt template and each line one model, standardized to its own 1987 to 2022 mean and standard deviation. The dashed line marks late 2022. For each combination we fit a straight line through its yearly scores. The slope of that line is the average change in z per year. Before 2022 the slopes point in no consistent direction, nine of the 18 upward and nine downward, and none is steeper than 0.08 z per year, which is small against a year-to-year spread of about 4 z . After 2022 all 18 point the same way, rising by between 1.24 and 3.94 z . The model-averaged series in Figure 5 does not show the pre-2022 level differences or the threefold spread in the size of the rise, which follows from the differing baseline standard deviations.

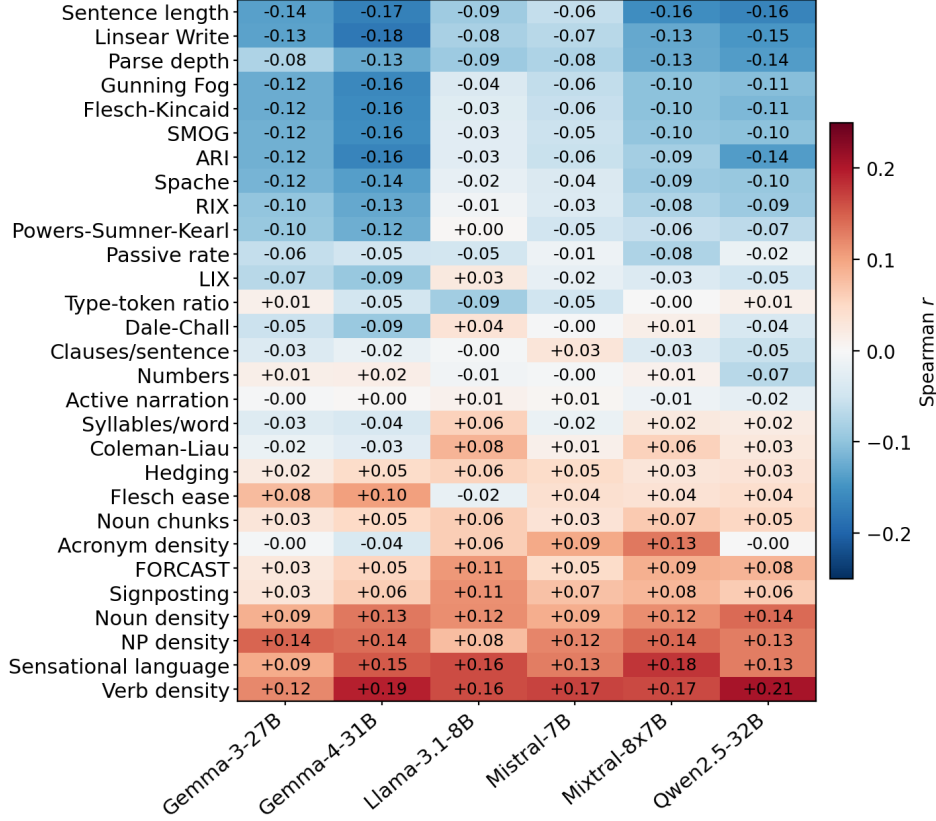


Figure 7: **Per-paper correlation between writing metrics and judge score.** Spearman correlation between each writing metric and each model’s judge score, over 24,754 NeurIPS abstracts. A model’s score for an abstract is the median of its nine generations, three prompts by three sampled runs. Rows are sorted by the correlation with the mean of those six per-model scores (1987–2024, the years the judge was scored). The judge score runs 1 to 5 with higher meaning easier, so each cell’s sign says which way that feature moves the judge. **Red (positive): more of that feature is judged *more* readable**, as for verb density, sensational language and noun phrase density. **Blue (negative): more of it is judged *less* readable**, as for sentence length and syntactic complexity, which is the direction the classical formulas also take. Near-white cells, including acronym density, indicate near indifference. All values are Spearman correlations and every effect is small ($|r| < 0.25$).